

Conflicting roles of conductive additives in controlling cathode performance in all-solid-state batteries

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ABSTRACT

Sulfide-based all-solid-state batteries (ASSBs) have recently attracted significant attention owing to the high ionic conductivities and mechanical ductilities of sulfide solid electrolytes (SEs). In general, carbon-based conductive additives (CAs) are incorporated into solid composite electrodes to enable facile electronic transport and to enhance active-material (AM) utilization. Herein, we reveal the conflicting roles of the one-dimensional (1D) CA (vapor-grown carbon fibers) in determining the electrochemical performance of composite electrodes with high AM fraction (f_{AM}) (i.e., low SE fraction) based on impedance decoupling analyses. The CA provides a beneficial effect on the performance of the low- f_{AM} electrode ($f_{AM} = 72$ wt%) by reducing its electronic resistance, whereas the CA-incorporated high- f_{AM} electrode ($f_{AM} = 88$ wt%) shows inferior rate-capability and severe capacity decay compared to the CA-free electrode. In-depth impedance decoupling analyses indicate that in high- f_{AM} electrodes with high CA-to-SE ratios, the CA makes the ionic pathway tortuous and accelerates the formation of SE-derived resistive phases, thus nullifying the benefits of enhanced electronic transport. In addition to the construction of optimized charge transport pathways, this study highlights the importance of the compatibility between the CA and SE, which is experimentally demonstrated by high- f_{AM} electrodes with halide-type SEs.

1. Introduction

All-solid-state batteries (ASSBs) have been regarded as promising next-generation energy storage systems due to their enhanced safety features and cell designs that can achieve higher energy density than conventional Li-ion batteries [1–4]. Over the past few decades, various types of solid electrolyte (SE), such as sulfides [5–7], oxides [8–10], and halides [11–13], have been extensively studied. Among various SE candidates, sulfide-based SEs, e.g., $\text{Li}_6\text{PS}_5\text{Cl}$ (LPSCl) and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$, have gained widespread attention due to their high ionic conductivities at room temperature ($> 1 \text{ mS cm}^{-1}$) and good mechanical processabilities [14,15]. Recently, such halide-type Li^+ conductors as Li_2ZrCl_6 (LZC) and Li_3InCl_6 have emerged as alternative SEs to sulfides because they are highly stable toward electrochemical oxidation ($> 4 \text{ V vs. Li/Li}^+$) [16,17].

Composite electrodes for ASSBs are typically composed of active materials (AMs), SEs, conductive additives (CAs), and binders, and their electrochemical properties are strongly dependent on the contents (fractions) and distributions of the constituents [18–20]. CAs are used to construct percolated electronic pathways among AM particles

throughout the electrode [21,22]. For conventional Li-ion batteries, AM and CA powders are first mixed with a polymeric binder in a solvent, and then, the resulting slurry is coated on a current collector, followed by the impregnation of a liquid electrolyte. For ASSBs, on the other hand, CA powders are mixed and pressed together with AM and SE particles (and binder) during fabrication. Therefore, care is needed when selecting and incorporating the CA into the electrode because CA particles can disconnect the percolated SE networks, thereby blocking the ionic transport [23].

Furthermore, poor electrochemical stability of sulfide SEs are known to cause unwanted side reactions with CAs [24]. Several experimental studies have shown that the incorporation of zero-dimensional (0D) carbon blacks deteriorates the electrochemical performance of sulfide-based ASSBs, depending on their contents in composite electrodes [25–28]. The electronically conducting CAs promote the oxidative decomposition of sulfide SEs during the charging process, which results in the accumulation of SE-derived resistive species within the electrodes [29,30]. Ates et al. reported that micro-sized 1D vapor-grown carbon fibers (VGCFs) offer advantages over conventional carbon blacks; that is, improved interparticle connectivity, and more

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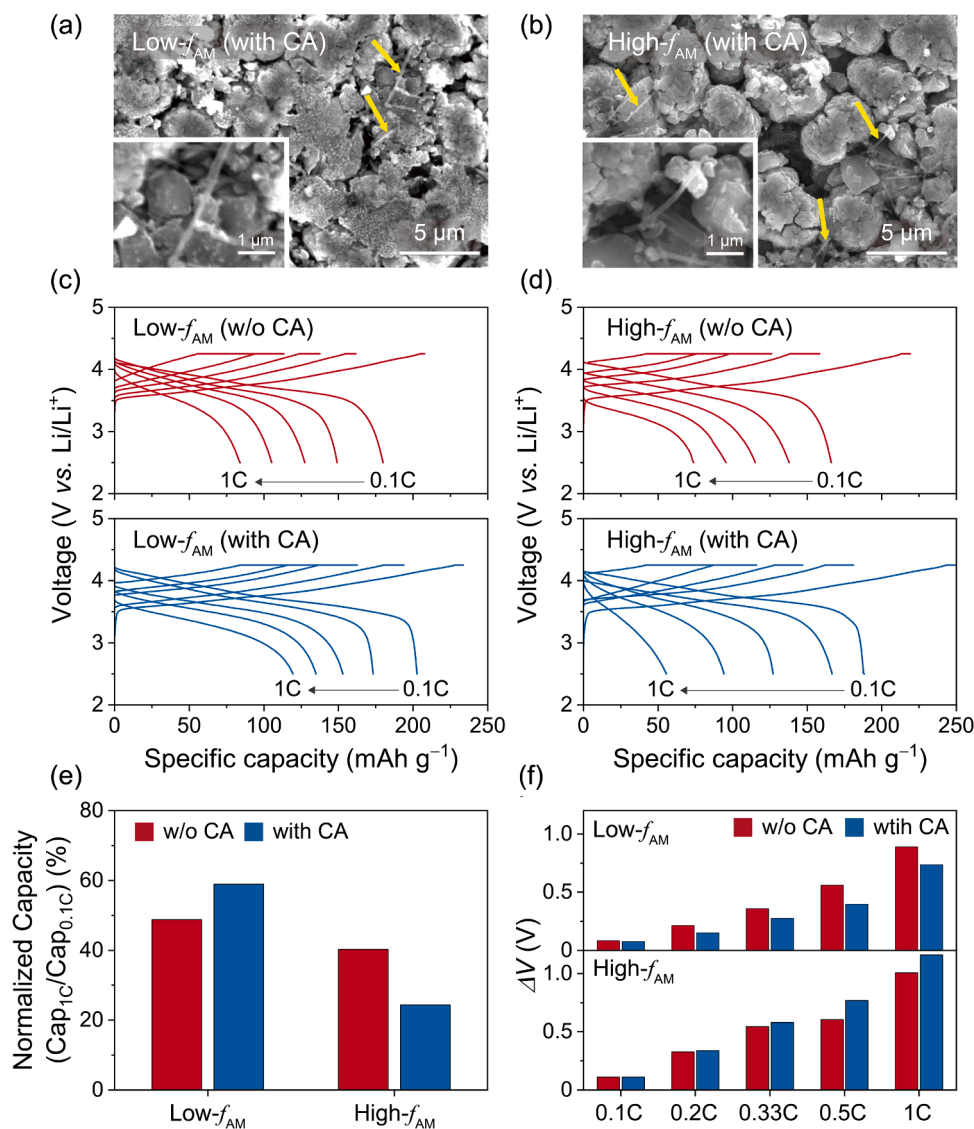


Fig. 1. SEM images of CA-incorporated (a) low- f_{AM} and (b) high- f_{AM} electrodes. Arrows exemplarily highlight VGCFs used as the CA. Voltage profiles of ASSBs using (c) low- f_{AM} and (d) high- f_{AM} electrodes at different rates of 0.1C–1.0C. (e) Comparison of the discharge capacity retained at 1C with respect to the capacity at 0.1C and (f) charge–discharge voltage gaps at SOC = 50 %.

importantly, enhanced interfacial stability with sulfide SEs owing to their low surface areas and reduced amounts of surface defects [31].

In this study, we investigated the roles played by the CA in determining the electrochemical performance of composite cathodes with different AM fractions (f_{AM}) on the basis of AC-impedance decoupling. VGCFs were employed as the CA for composite cathodes comprising LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ (NCM) and LPSCl. The CA-incorporated low- f_{AM} electrode showed improved electrochemical performance as compared with the CA-free electrode, whereas the CA-incorporated high- f_{AM} electrode exhibited poor rate-capability and accelerated capacity decay with significant increases in ionic and interfacial resistances during cycling. In addition to the formation of tortuous ionic pathways, the critical degradation factor of the CA-incorporated high- f_{AM} electrode was the accumulation of SE-derived resistive species at the three-phase interfaces among AM, CA, and SE particles, which could be improved by using a halide-type LZC SE.

2. Experimental

2.1. Electrode preparation

Polycrystalline NCM and LPSCl were supplied by COSMO AM&T (Korea). LZC powder was prepared by subjecting a stoichiometric mixture comprising LiCl (Sigma Aldrich, 99.9 %) and ZrCl₄ (Sigma Aldrich, 99.9 %) to high-energy ball milling at 600 rpm for 3 h, followed by annealing at 260 °C for 12 h. The ionic conductivities of LPSCl and LZC were measured to 4.8 and 0.6 mS cm⁻¹, respectively (Figs. S1 and S2). VGCFs were obtained from Sigma Aldrich. NCM and the VGCFs were dried overnight in a vacuum oven at 200 °C. Composite cathodes were prepared by mixing NCM, LPSCl, and the VGCFs using a ball mill. The f_{AM} value was either 72 or 88 wt%, and 2 wt% VGCFs were added to the composite electrode.

2.2. Material characterization

The microstructures and morphologies of materials and electrodes were investigated via field-emission scanning electron microscopy (FE-

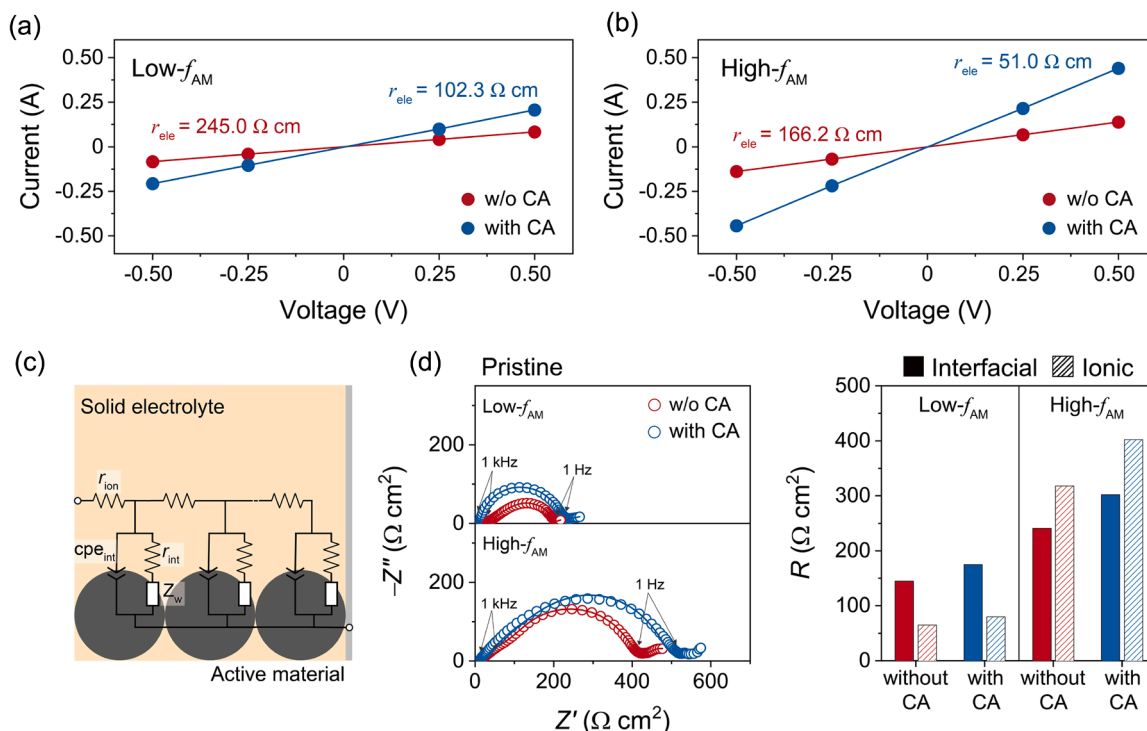


Fig. 2. Steady-state cell currents at various DC polarization potentials for (a) low- f_{AM} and (b) high- f_{AM} electrodes. (c) TLM equivalent circuit used to fit the AC-impedance spectra. (d) AC-impedance spectra of pristine composite electrodes along with the ionic and interfacial resistances determined by the fitting.

SEM, Verios G4UC). The porous and structural properties of VGCFs were characterized via N_2 adsorption-desorption measurements (AutoChem 2950) and Raman spectroscopy (Nicolet Almega XR, Thermo Scientific). X-ray photoelectron spectroscopy (XPS) with a monochromatic Al K_{α} source (1486.6 eV) and a K-Alpha⁺ spectrometer (Thermo Fisher Scientific) was used to determine the chemical state of each element.

2.3. ASSB assembly

ASSBs were assembled using a poly(aryl-ether-ether-ketone) (PEEK) mold with a hole and stainless-steel rods (diameter = 10 mm). LPSCI powder (100 mg) was first poured into the PEEK mold and gently pressed. The composite cathode mixture (10 mg, diameter = 10 mm) was then added onto the SE layer and pressed at 433 MPa, after which In and Li discs were added to the other side of the SE layer. The AM loadings of the composite cathodes with $f_{AM} = 72$ and 88 wt% are 9.17 and 11.21 mg cm^{-2} , respectively. The cells were operated at a stack pressure of 250 MPa following fabrication. For cyclic voltammetry (CV) measurements to examine the SE/CA compatibility, LPSCI and VGCF powders were mixed in volume ratios of 94:6 and 79:21, and SUS and Li-In foils were used as the working and counter electrodes, respectively.

2.4. Electrochemical measurements

DC-polarization experiments were performed to determine the electronic resistance of the composite electrode using an ion-blocking [stainless-steel | composite electrode | stainless-steel] cell. ASSB cells were subjected to galvanostatic cycling in the 2.5–4.25 V vs. Li/Li⁺ voltage range at 30 °C. Initial cycling was performed at 0.1C with constant current-constant voltage (CC–CV) and constant current (CC) modes used for charging and discharging, respectively. The CV cut-off current was set to 20 % of the applied current in CC mode. Cells were charged and discharged at 0.33C following the initial cycle. Rate-capabilities were determined at various C-rates of 0.1–1.0C. AC-impedance spectroscopy was performed to determine the ionic and

interfacial resistances using an impedance analyzer (Biologic SP-300) in the 7 MHz to 30 mHz frequency range with an AC amplitude of 5 mV. CV was performed in the 2.5–4.3 V vs. Li/Li⁺ voltage range at 10 mV s^{−1}.

3. Results and discussion

To investigate how the CA affects the ASSB performance, CA-incorporated and CA-free NCM-LPSCI composite electrodes were prepared. Micro-sized VGCFs with high aspect ratios (Fig. S3) and small specific surface area (12.3 m² g^{−1}, Fig. S4) were used as the CA. The SEM images of the composite cathodes obtained after cell fabrication are shown in Figs. 1a and b and S5. The composite cathodes exhibited homogeneous distributions of NCM particles for $f_{AM} = 72$ and 88 wt% as well. Fig. 1c and d present the voltage profiles of ASSBs with low- and high- f_{AM} electrodes at different C-rates ranging from 0.1C to 1C. The CA-incorporated low- f_{AM} electrode delivered significantly higher discharge capacities, compared to the CA-free electrode (Fig. 1c), which indicates the beneficial role of the CA in enhancing the electrode performance. As shown in Fig. 1e, the difference in the discharge capacity was apparent at 1C: 119.5 mAh g^{−1} for the CA-incorporated electrode versus 84.2 mAh g^{−1} for the CA-free electrode. By contrast, the CA-incorporated high- f_{AM} electrode showed an initial Coulombic efficiency as low as 72.4 % and exhibited a considerably reduced capacity with an increasing C-rate (Figs. 1d and e). Fig. 1f compares the voltage gap (ΔV) on charge-discharge curves at a state of charge (SOC) = 50 %. The ΔV value was decreased by the incorporation of the CA into the low- f_{AM} electrode ($\Delta V = 0.88$ V (CA-free) → 0.73 V (CA-incorporated) at 1C), whereas the CA-incorporated high- f_{AM} electrode exhibited increased polarizations ($\Delta V = 1.09$ V (CA-free) → 1.29 V (CA-incorporated) at 1C).

To identify the primary factors responsible for the differences in rate-performance, composite electrodes were subjected to DC polarization (Fig. 2a and b) and AC-impedance measurements (Figs. 2c–e). Fig. 2a and b show the steady-state currents measured at various DC polarization potentials. The electronic resistance per unit length (r_{ele}) was determined based on the Ohm's law to conduct the quantitative

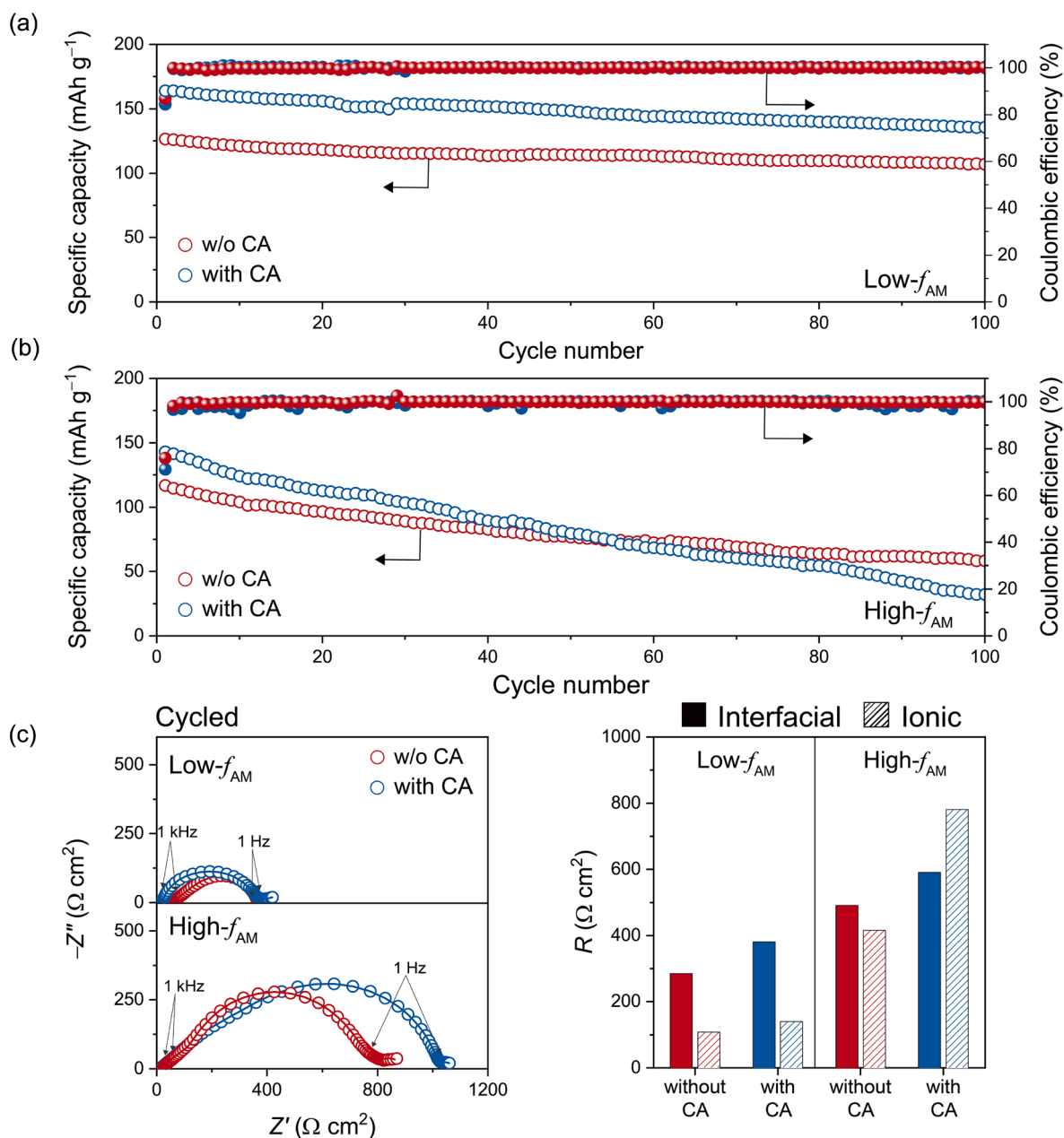


Fig. 3. Capacity retentions and Coulombic efficiencies of ASSBs with (a) low- f_{AM} and (b) high- f_{AM} electrodes. (c) AC-impedance spectra of the cycled composite electrodes along with the ionic and interfacial resistances determined by the fitting.

evaluation of electronic conduction in the composite electrode. The incorporation of CAs in both low- and high- f_{AM} electrodes resulted in a sharp decrease in r_{ele} , which verifies the effectiveness of the CA in forming the extended electronic networks. The beneficial effect of the CA in facilitating electronic transport was more evident in the high- f_{AM} electrode, which suggests that the kinetics of electronic transport is not responsible for the poor rate-capability of the CA-incorporated high- f_{AM} electrode.

Fig. 2c illustrates an equivalent circuit based on a modified transmission-line model (TLM) used to fit the collected AC-impedance spectra. TLMs have been widely used to describe the electrochemical behavior of porous electrodes with cylindrical pores. In this study, the TLM was modified to describe the multiple reaction processes in a solid composite cathode. The circuit in Fig. 2c is expressed by a network of distributed impedance elements, which represent (i) ionic transport in the SE phase, (ii) charge transfer and double layer charging at SE/NCM

interfaces, and (iii) solid-state diffusion in NCM particles [32,33]. It was assumed that the electronic conductivity of the composite cathode is sufficiently high. In the circuit, r_{ion} denotes the resistance per unit length (Ω cm) for the ionic transport. The interfacial impedance consists of r_{ion} , cpe_{int} , and z_w , which correspond to the interfacial resistance, constant-phase element, and Warburg impedance, respectively. The measured impedance spectra of the composite cathodes in Fig. 2d were fitted to the modified TLM, and the fitted parameters are listed in Table S1.

Overall, the high- f_{AM} electrodes had larger R_{ion} values than R_{int} , regardless of the absence and presence of the CA, whereas the low- f_{AM} electrode showed the opposite behavior. The area-specific ionic resistance (R_{ion}) and interfacial resistance (R_{int}) values were slightly higher for the CA-incorporated low- f_{AM} electrode in comparison with the CA-free electrode. On the other hand, the incorporation of CA in the high- f_{AM} electrode considerably increased both ionic and interfacial

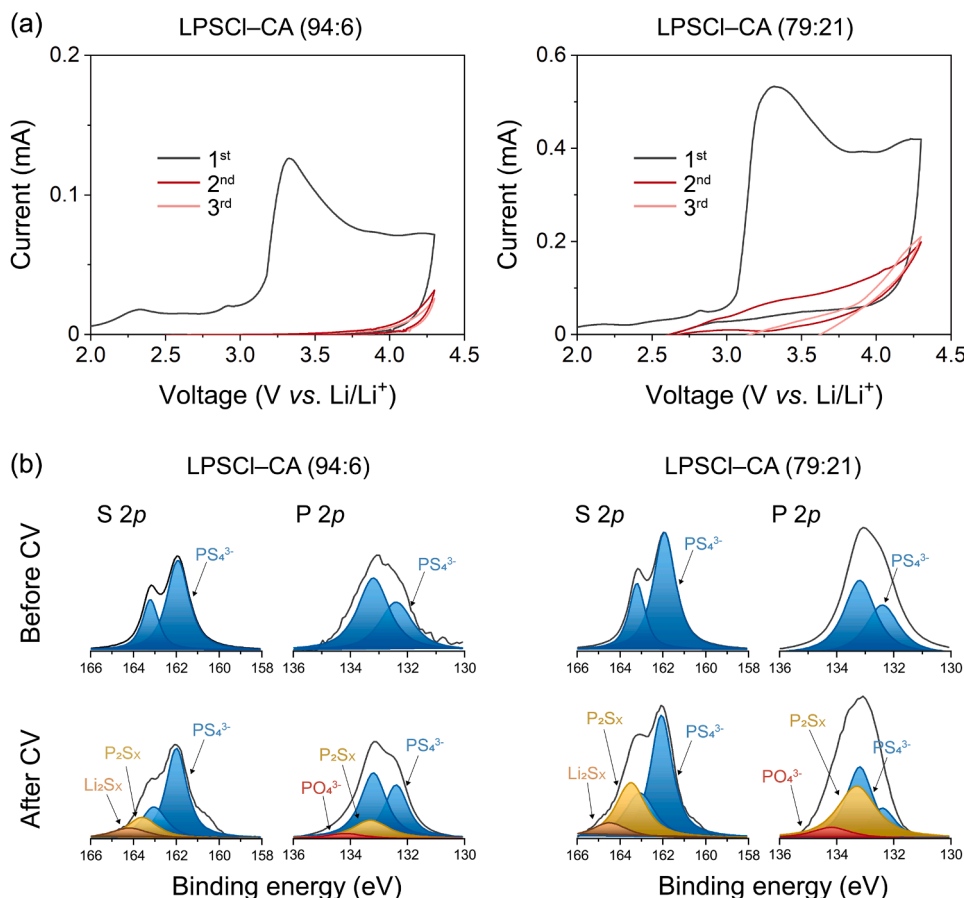


Fig. 4. (a) Cyclic voltammograms of LPSCI–CA composites. (b) XPS spectra for S 2p and P 2p regions obtained from the LPSCI–CA composites before and after CV measurements.

resistances, which could account for the inferior rate-capability observed in Fig. 1. Such changes in R_{ion} ($318 \Omega \text{ cm}^2$ (CA-free) $\rightarrow 402 \Omega \text{ cm}^2$ (CA-incorporated)) and R_{int} ($241 \Omega \text{ cm}^2$ (CA-free) $\rightarrow 278 \Omega \text{ cm}^2$ (CA-incorporated)) imply that the CA hinders facile ionic transport through the SE phase and interfacial charge transfer between NCM and LPSCI in the high- f_{AM} electrode. In fact, the tortuosity values of high- f_{AM} electrodes were remarkably higher than those of low- f_{AM} electrodes, and the tortuosity was further increased by the incorporation of the CA (Table S2) [34].

Fig. 3a and b compare the long-term cycling performance of low- and high- f_{AM} electrodes, respectively. The corresponding voltage profiles are depicted in Figs. S6 and S7. The cells were cycled between 2.5 and 4.25 V vs. Li/Li⁺ at 0.33C after an initial cycle at 0.1C. As shown in Fig. 3a, the CA-incorporated low- f_{AM} electrode exhibited higher discharge capacities compared to the CA-free electrode and maintained similar capacity retentions over 100 cycles. This confirms the effectiveness of the CA for improving the AM utilization by forming facile electronic conduction networks. On the other hand, the CA-incorporated high- f_{AM} electrode showed a higher initial discharge capacity than that of the CA-free electrode, but it exhibited more rapid capacity decay during cycling. The impedance signals measured from low- and high- f_{AM} electrodes after cycling are presented in Fig. 3c. The impedance spectra were fitted to the equivalent circuit in Fig. 2c, and the resulting parameters are listed in Table S1. It is noted from Figs. 2d and 3c that the CA-incorporated high- f_{AM} electrode showed steep increases in R_{ion} ($402 \rightarrow 781 \Omega \text{ cm}^2$) and R_{int} ($278 \rightarrow 591 \Omega \text{ cm}^2$) after 100 cycles, indicating that the observed capacity loss originated mainly from the degraded ionic pathways and interfaces.

To understand the distinct impedance characteristics of low- and high- f_{AM} electrodes with CAs, the reactivity between LPSCI and CA was

investigated by CV measurements. Given the weight fractions of LPSCI and CA in low- and high- f_{AM} electrodes, two types of LPSCI–CA composites with different volume ratios were fabricated: (1) 94:6 and (2) 79:21. As shown in Fig. 4a, the oxidative current of composite (2) was higher than that of composite (1) over the entire voltage range, which can be attributed to the higher contact areas between the CA and LPSCI. Because the high current during the anodic scan is related to the increased decomposition of LPSCI [35], it can be inferred that the CA-incorporated high- f_{AM} electrode suffered from more severe degradation of LPSCI/CA interfaces compared to the low- f_{AM} electrode. The current decreased for the next two cycles for both composites (1) and (2), but composite (2) underwent interfacial degradation even after the serious decomposition of LPSCI upon the first cycle, as evidenced by high oxidative currents in the subsequent cycles.

Further experimental evidence of the side reaction between the CA and LPSCI was acquired by XPS. Fig. 4b compares the XPS spectra of S 2p and P 2p core levels in the LPSCI–CA composites before and after the CV tests. The S 2p spectra of pristine LPSCI in both composites showed characteristic peaks at 161.7 and 163.0 eV corresponding to the PS₄³⁻ unit of argyrodites [36,37]. After CV cycling, new signals appeared at 163.7 and 164.2 eV, which can be assigned to P₂S_x and Li₂S_n, respectively [38,39]. The PO₄³⁻ peak at 134.1 eV was also attributed to the oxidized species formed by the side reaction of LPSCI with oxygen-containing species on the CA surface (–OH groups, carbonates, and water residuals) [40–42]. The XPS results indicated that composite (2) had a much larger amount of decomposition products compared to composite (1) after CV cycling. The CV and XPS analyses demonstrate that the CA causes the significant decomposition of LPSCI in the high- f_{AM} electrode, and thus, the SE-derived resistive phases can make the ionic transport and interfacial reaction sluggish, which explains the high R_{ion}

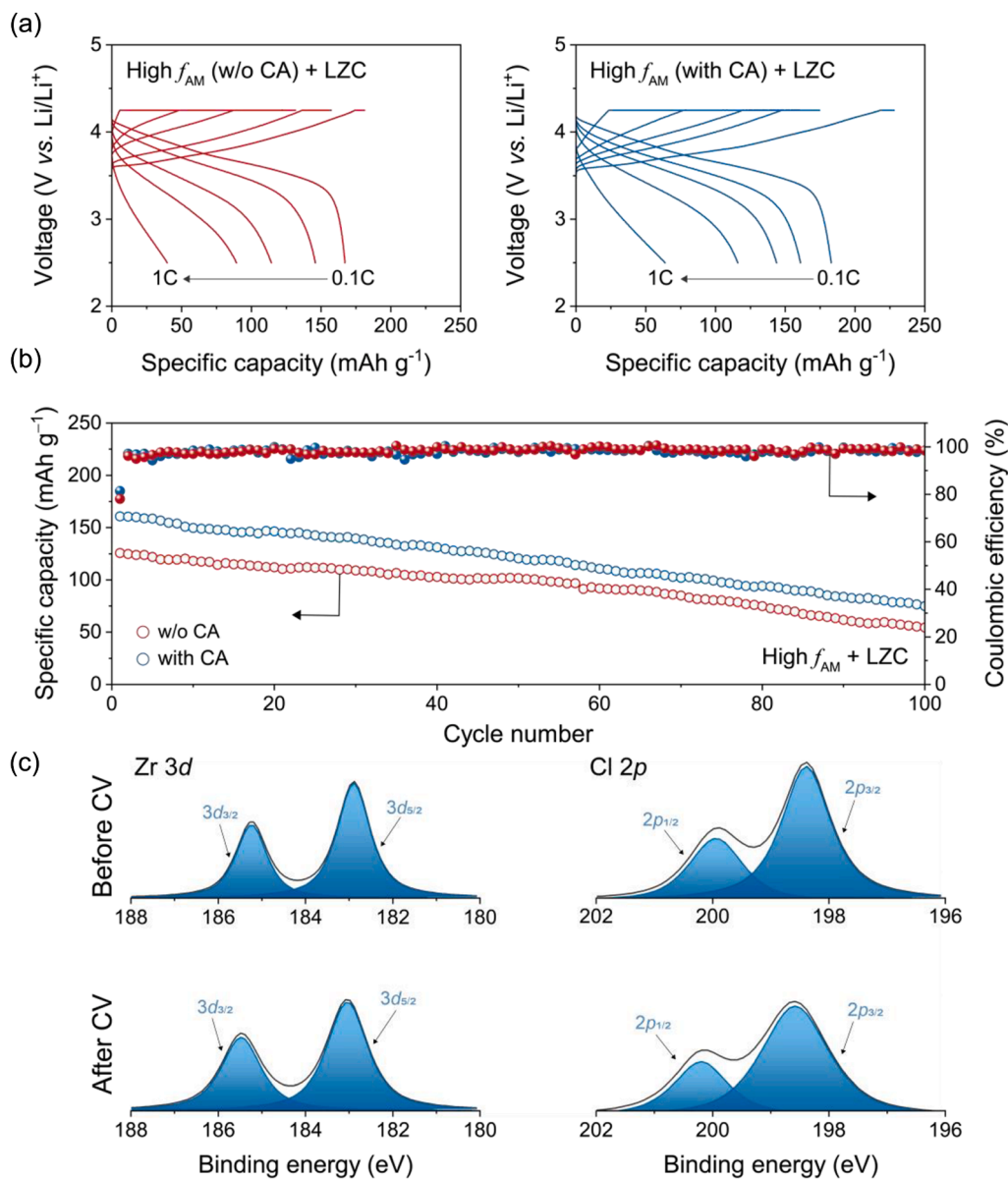


Fig. 5. (a) Rate-capability and (b) cycling performance of CA-free and CA-incorporated high- f_{AM} electrodes with LZC. (c) XPS spectra for Zr 3d and Cl 2p regions obtained from the LZC–CA composites before and after CV measurements.

and R_{int} values of the cycled electrode in Fig. 3c.

The electrochemical and post-mortem analyses clearly showed that the increases in R_{ion} and R_{int} and the resultant performance degradation arise from the accumulation of LPSCI-derived resistive products within the electrode. In addition, the negative role played by the CA in degrading the performance is more evident in the high- f_{AM} electrode than in the low- f_{AM} electrode. The importance of the compatibility between the CA and SE was further demonstrated by the high- f_{AM} electrode with the halide-type LZC SE. LZC is known to be more electrochemically stable at high voltages, compared to sulfide-based SEs [17,43]. Fig. 5a compares the rate-performance of high- f_{AM} electrodes with LZC instead of LPSCI. In contrast to Fig. 1d, the CA-incorporated high- f_{AM} electrode exhibited higher discharge capacities over the whole C-rate range, which can be explained in terms of the enhanced kinetics of electronic transport (see the electronic resistance measurements in Fig. S8). Furthermore, its cycling behavior over 100 cycles was quite similar to that of the CA-free electrode (Figs. 5b and S9). The XPS spectra of the CV-cycled LZC–CA composite in Fig. 5c showed no signs of significant side reactions between the CA and LZC. Although the ionic conductivity

of LZC should be improved for practical applications, the electrochemical test results using LZC suggest that CAs can play a beneficial role in improving the rate-performance and cyclability of high- f_{AM} electrodes when coupled with electrochemically stable SEs.

Fig. 6 schematically illustrates the roles of the CA in the electrochemical performance of high- f_{AM} electrodes. In general, the electronic transport is limited in the absence of CAs, and the incorporation of CAs into low- f_{AM} electrodes lead to enhanced electronic transport and AM utilization. In high- f_{AM} electrodes with high CA-to-SE ratios, however, CAs not only disconnect the ionic transport networks but also promote the SE decomposition; as a result, the SE-derived resistive phases make the ionic pathway more tortuous and further inhibit the charge transfer reactions, thus nullifying the benefits of enhanced electronic transport. The use of CAs with SEs with high conductivity and electrochemical stability provides an effective approach to enhancing both electronic/ionic transport and interfacial kinetics.

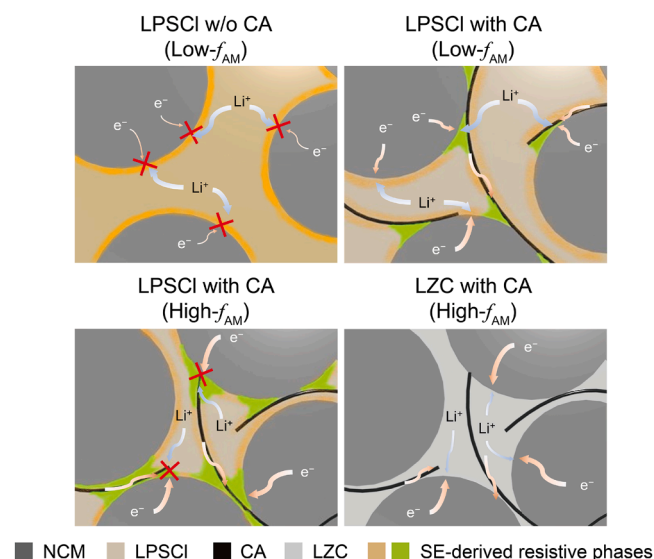


Fig. 6. Schematic illustrations of the roles of CAs in controlling multiple reaction kinetics in composite cathodes for ASSBs.

4. Conclusion

We investigated the role of 1D-type CAs (VGCFs) in determining the performance of solid composite electrodes with different AM fractions by AC-impedance decoupling combined with CV and XPS analyses. The low- f_{AM} electrode showed the improved electrochemical performance in the presence of the CA, while the CA-incorporated high- f_{AM} electrode exhibited the reduced rate-performance and cycling stability as compared with the CA-free electrode. The impedance signals decoupled using the TLM revealed that the CA incorporation led to the considerable increases in ionic and interfacial resistances of the high- f_{AM} electrode. The combined electrochemical and post-mortem analyses demonstrated that the CA disconnects the ionic transport networks and promote the SE decomposition; therefore, the SE-derived resistive phases further leads to the formation of the tortuous ionic pathways and inhibit the intercalation kinetics at the AM/SE interfaces, while causing the enhanced electronic transport to have no beneficial effect. This study conveys an important message that in addition to the construction of optimized electronic/ionic pathways, the compatibility between the CA and SE should be taken into account when designing high-performance electrodes, which was demonstrated by high- f_{AM} electrodes with LZC SEs.

CRedit authorship contribution statement

Minhyeong Cho: Writing – original draft, Investigation. **Jonghyeok Yun:** Investigation. **Junhee Kang:** Investigation. **Siwon Kim:** Investigation. **Jong-Won Lee:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.electacta.2024.143990](https://doi.org/10.1016/j.electacta.2024.143990).

References

- [1] W.Z. Huang, C.Z. Zhao, P. Wu, H. Yuan, W.E. Feng, Z.Y. Liu, Y. Lu, S. Sun, Z.H. Fu, J.K. Hu, S.J. Yang, J.Q. Huang, Q. Zhang, Anode-free solid-state lithium batteries: a review, *Adv. Energy Mater.* 12 (2022) 2201044, <https://doi.org/10.1002/aenm.202201044>.
- [2] W. Jeong, S.S. Park, J. Yun, H.R. Shin, J. Moon, J.W. Lee, Tailoring grain boundary structures and chemistry of $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ solid electrolytes for enhanced air stability, *Energy Storage Mater.* 54 (2023) 543–552, <https://doi.org/10.1016/j.ensm.2022.10.044>.
- [3] Y.M. Jin, Q.S. He, G.Z. Liu, Z. Gu, M. Wu, T.Y. Sun, Z.H. Zhang, L.F. Huang, X. Y. Yao, Fluorinated $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ enables stable all-solid-state lithium batteries, *Adv. Mater.* 35 (2023) 2211047, <https://doi.org/10.1002/adma.202211047>.
- [4] E.S. Won, H.R. Shin, W. Jeong, J. Yun, J.W. Lee, Biphasic solid electrolytes with homogeneous Li-ion transport pathway enabled by metal-organic frameworks, *Electrochim. Acta* 418 (2022) 140374, <https://doi.org/10.1016/j.electacta.2022.140374>.
- [5] N. Kamaya, K. Homma, Y. Yamakawa, M. Hirayama, R. Kanno, M. Yonemura, T. Kamiyama, Y. Kato, S. Hama, K. Kawamoto, A. Mitsui, A lithium superionic conductor, *Nat. Mater.* 10 (2011) 682–686, <https://doi.org/10.1038/Nmat3066>.
- [6] Y. Kato, S. Hori, T. Saito, K. Suzuki, M. Hirayama, A. Mitsui, M. Yonemura, H. Iba, R. Kanno, High-power all-solid-state batteries using sulfide superionic conductors, *Nat. Energy* 1 (2016) 16030, <https://doi.org/10.1038/Nenergy.2016.30>.
- [7] J.Y. Yin, X.Y. Yao, G. Peng, J. Yang, Z. Huang, D. Liu, Y.C. Tao, X.X. Xu, Influence of the Li-Ge-P-S based solid electrolytes on NCA electrochemical performances in all-solid-state lithium batteries, *Solid State Ion.* 274 (2015) 8–11, <https://doi.org/10.1016/j.ssi.2015.02.014>.
- [8] A.J. Samson, K. Hofstetter, S. Bag, V. Thangadurai, A bird's-eye view of Li-stuffed garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ceramic electrolytes for advanced all-solid-state Li batteries, *Energy Environ. Sci.* 12 (2019) 2957–2975, <https://doi.org/10.1039/c9ee01548e>.
- [9] C.W. Wang, K. Fu, S.P. Kammampata, D.W. Mcowen, A.J. Samson, L. Zhang, G. T. Hitz, A.M. Nolan, E.D. Wachsman, Y.F. Mo, V. Thangadurai, L.B. Hu, Garnet-type solid-state electrolytes: materials, interfaces, and batteries, *Chem. Rev.* 120 (2020) 4257–4300, <https://doi.org/10.1021/acs.chemrev.9b00427>.
- [10] Y.C. Kim, K.N. Jung, J.W. Lee, M.S. Park, Improving the ionic conductivity of $\text{Li}_{1-x}\text{Al}_x\text{Ge}_{2-x}(\text{PO}_4)_3$ solid electrolyte for all-solid-state batteries using microstructural modifiers, *Ceram. Int.* 46 (2020) 23200–23207, <https://doi.org/10.1016/j.ceramint.2020.06.101>.
- [11] T. Asano, A. Sakai, S. Ouchi, M. Sakaida, A. Miyazaki, S. Hasegawa, Solid halide electrolytes with high lithium-ion conductivity for application in 4V class bulk-type all-solid-state batteries, *Adv. Mater.* 30 (2018) 1803075, <https://doi.org/10.1002/adma.201803075>.
- [12] X.N. Li, J.W. Liang, J. Luo, M.N. Banis, C.H. Wang, W.H. Li, S.X. Deng, C. Yu, F. P. Zhao, Y.F. Hu, T.K. Sham, L. Zhang, S.Q. Zhao, S.G. Lu, H. Huang, R.Y. Li, K. R. Adair, X.L. Sun, Air-stable Li_3InCl_6 electrolyte with high voltage compatibility for all-solid-state batteries, *Energy Environ. Sci.* 12 (2019) 2665–2671, <https://doi.org/10.1039/c9ee02311a>.
- [13] Q.Y. Luo, C. Yu, C.C. Wei, S. Chen, S.Q. Chen, Z.L. Jiang, L.F. Peng, S.J. Cheng, J. Xie, Enabling superior electrochemical performances of $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$ -based all-solid-state batteries using lithium halide electrolytes, *Ceram. Int.* 49 (2023) 11485–11493, <https://doi.org/10.1016/j.ceramint.2022.11.347>.
- [14] J.H. Wu, S.F. Liu, F.D. Han, X.Y. Yao, C.S. Wang, Lithium/sulfide all-solid-state batteries using sulfide electrolytes, *Adv. Mater.* 33 (2021) 2000751, <https://doi.org/10.1002/adma.202000751>.
- [15] S.J. Chen, D.J. Xie, G.Z. Liu, J.P. Mwisirwa, Q. Zhang, Y.R. Zhao, X.X. Xu, X. Y. Yao, Sulfide solid electrolytes for all-solid-state lithium batteries: structure, conductivity, stability and application, *Energy Storage Mater.* 14 (2018) 58–74, <https://doi.org/10.1016/j.ensm.2018.02.020>.
- [16] L.D. Zhou, T.T. Zuo, C.Y. Kwok, S.Y. Kim, A. Assoud, Q. Zhang, J. Janek, L. F. Nazar, High areal capacity, long cycle life 4V ceramic all-solid-state Li-ion batteries enabled by chloride solid electrolytes, *Nat. Energy* 7 (2022) 83–93, <https://doi.org/10.1038/s41560-021-00952-0>.
- [17] Q.N. Shao, C.H. Yan, M.X. Gao, W.B. Du, J. Chen, Y.X. Yang, J.T. Gan, Z.J. Wu, W. P. Sun, Y.Z. Jiang, Y.F. Liu, M.X. Gao, H.G. Pan, New insights into the effects of Zr substitution and carbon additive on $\text{Li}_{3-x}\text{Er}_{1-x}\text{Zr}_x\text{Cl}_6$ halide solid electrolytes, *ACS Appl. Mater. Interfaces* 14 (2022) 8095–8105, <https://doi.org/10.1021/acsaami.1c25087>.
- [18] T. Shi, Q.S. Tu, Y.S. Tian, Y.H. Xiao, L.J. Miara, O. Kononova, G. Ceder, High active material loading in all-solid-state battery electrode via particle size optimization,

- Adv. Energy Mater. 10 (2020) 1902881, <https://doi.org/10.1002/aenm.201902881>.
- [19] A. Bielefeld, D.A. Weber, J. Janek, Microstructural modeling of composite cathodes for all-solid-state batteries, *J. Phys. Chem. C* 123 (2019) 1626–1634, <https://doi.org/10.1021/acs.jpcc.8b11043>.
- [20] P. Minnmann, L. Quillman, S. Burkhardt, F.H. Richter, J. Janek, Editors' choice—quantifying the impact of charge transport bottlenecks in composite cathodes of all-solid-state batteries, *J. Electrochem. Soc.* 168 (2021) 040537, <https://doi.org/10.1149/1945-7111/abf8d7>.
- [21] M.E. Spahr, D. Goers, A. Leone, S. Stallone, E. Grivei, Development of carbon conductive additives for advanced lithium ion batteries, *J. Power Sources* 196 (2011) 3404–3413, <https://doi.org/10.1016/j.jpowsour.2010.07.002>.
- [22] V. Palomares, A. Goñi, I.G. de Muro, I. de Meatza, M. Bengoechea, I. Cantero, T. Rojo, Conductive additive content balance in Li-ion battery cathodes: commercial carbon blacks vs. carbon from LiFePO_4/C composites, *J. Power Sources* 195 (2010) 7661–7668, <https://doi.org/10.1016/j.jpowsour.2010.05.048>.
- [23] F.Y. Su, Y.B. He, B.H. Li, X.C. Chen, C.H. You, W. Wei, W. Lv, Q.H. Yang, F.Y. Kang, Could graphene construct an effective conducting network in a high-power lithium ion battery? *Nano Energy* 1 (2012) 429–439, <https://doi.org/10.1016/j.nanoen.2012.02.004>.
- [24] J. Lau, R.H. DeBlock, D.M. Butts, D.S. Ashby, C.S. Choi, B.S. Dunn, Sulfide solid electrolytes for lithium battery applications, *Adv. Energy Mater.* 8 (2018) 1800933, <https://doi.org/10.1002/aenm.201800933>.
- [25] W.B. Zhang, T. Leichtweiss, S.P. Culver, R. Koerver, D. Das, D.A. Weber, W. G. Zeier, J. Janek, The detrimental effects of carbon additives in $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ -based solid-state batteries, *ACS Appl. Mater. Interfaces* 9 (2017) 35888–35896, <https://doi.org/10.1021/acsami.7b11530>.
- [26] S.X. Deng, Y.P. Sun, X. Li, Z.H. Ren, J.W. Liang, K. Doyle-Davis, J.N. Liang, W.H. Li, M.N. Banis, Q. Sun, R.Y. Li, Y.F. Hu, H. Huang, L. Zhang, S.G. Lu, J. Luo, X.L. Sun, Eliminating the detrimental effects of conductive agents in sulfide-based solid-state batteries, *ACS Energy Lett* 5 (2020) 1243–1251, <https://doi.org/10.1021/acsenergylett.0c00256>.
- [27] H.Q. Shen, S.H. Jing, S.L. Liu, Y.T. Huang, F.B. He, Y. Liu, Z. Zhuang, Z.L. Zhang, F. Y. Liu, Tailoring the electronic conductivity of high-loading cathode electrodes for practical sulfide-based all-solid-state batteries, *Adv. Powder Mater.* 2 (2023) 100136, <https://doi.org/10.1016/j.apmater.2023.100136>.
- [28] N. Lee, J. Lee, T. Lee, J. Oh, I. Hwang, G. Seo, H. Kim, J.W. Choi, Rationally designed solution-processible conductive carbon additive coating for sulfide-based all-solid-state batteries, *ACS Appl. Mater. Interfaces* 15 (2023) 34931–34940, <https://doi.org/10.1021/acsami.3c05713>.
- [29] K. Yoon, J.J. Kim, W.M. Seong, M.H. Lee, K. Kang, Investigation on the interface between $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ electrolyte and carbon conductive agents in all-solid-state lithium battery, *Sci. Rep.* 8 (2018) 10366, <https://doi.org/10.1038/s41598-018-28601-9>.
- [30] G. Oh, M. Hirayama, O. Kwon, K. Suzuki, R. Kanno, Bulk-type all solid-state batteries with 5V class $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$ cathode and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ solid electrolyte, *Chem. Mater.* 28 (2016) 2634–2640, <https://doi.org/10.1021/acs.chemmater.5b04940>.
- [31] T. Ates, M. Keller, J. Kulisch, T. Adermann, S. Passerini, Development of an all-solid-state lithium battery by slurry-coating procedures using a sulfidic electrolyte, *Energy Storage Mater* 17 (2019) 204–210, <https://doi.org/10.1016/j.ensm.2018.11.011>.
- [32] J. Yun, H.R. Shin, T.D. Hoang, S. Kim, J.H. Choi, B. Kim, H. Jung, J. Moon, J. W. Lee, Deciphering the critical degradation factors of solid composite electrodes with halide electrolytes: interfacial reaction versus ionic transport, *Energy Storage Mater* 59 (2023) 102787, <https://doi.org/10.1016/j.ensm.2023.102787>.
- [33] J. Kang, H.R. Shin, J. Yun, S. Kim, B. Kim, K. Lee, Y. Lim, J.W. Lee, Chemo-mechanical failure of solid composite cathodes accelerated by high-strain anodes in all-solid-state batteries, *Energy Storage Mater* 63 (2023) 103049, <https://doi.org/10.1016/j.ensm.2023.103049>.
- [34] S.Y. Park, J. Jeong, H.C. Shin, Geometrical effect of active material on electrode tortuosity in all-solid-state lithium battery, *Appl. Sci.* 12 (2022) 12692, <https://doi.org/10.3390/app122412692>.
- [35] S. Randau, F. Walther, A. Neumann, Y. Schneider, R.S. Negi, B. Mogwitz, J. Sann, K. Becker-Steinberger, T. Danner, S. Hein, A. Latz, F.H. Richter, J. Janek, On the additive microstructure in composite cathodes and alumina-coated carbon microwires for improved all-solid-state batteries, *Chem. Mater.* 33 (2021) 1380–1393, <https://doi.org/10.1021/acs.chemmater.0c04454>.
- [36] F. Walther, R. Koerver, T. Fuchs, S. Ohno, J. Sann, M. Rohnke, W.G. Zeier, J. Janek, Visualization of the interfacial decomposition of composite cathodes in argyrodite-based all-solid-state batteries using time-of-flight secondary-ion mass spectrometry, *Chem. Mater.* 31 (2019) 3745–3755, <https://doi.org/10.1021/acs.chemmater.9b00770>.
- [37] X.W. Sun, L.L. Wang, J. Ma, X.R. Yu, S. Zhang, X.H. Zhou, G.L. Cui, A bifunctional chemomechanics strategy to suppress electrochemo-mechanical failure of Ni-rich cathodes for all-solid-state lithium batteries, *ACS Appl. Mater. Interfaces* 14 (2022) 17674–17681, <https://doi.org/10.1021/acsami.2c02678>.
- [38] F. Auvergniot, A. Cassel, J.B. Ledeuil, V. Viallet, V. Seznec, R. Dedryvère, Interface stability of argyrodite $\text{Li}_6\text{PS}_5\text{Cl}$ toward LiCoO_2 , $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$, and LiMn_2O_4 in bulk all-solid-state batteries, *Chem. Mater.* 29 (2017) 3883–3890, <https://doi.org/10.1021/acs.chemmater.6b04990>.
- [39] D.H.S. Tan, E.A. Wu, H. Nguyen, Z. Chen, M.A.T. Marple, J.M. Dour, X.F. Wang, H. D. Yang, A. Banerjee, Y.S. Meng, Elucidating reversible electrochemical redox of $\text{Li}_6\text{PS}_5\text{Cl}$ solid electrolyte, *ACS Energy Lett* 4 (2019) 2418–2427, <https://doi.org/10.1021/acsenergylett.9b01693>.
- [40] S.W. Park, G. Oh, J.W. Park, Y.C. Ha, S.M. Lee, S.Y. Yoon, B.G. Kim, Graphitic hollow nanocarbon as a promising conducting agent for solid-state lithium batteries, *Small* 15 (2019) 1900235, <https://doi.org/10.1002/smll.201900235>.
- [41] T. Kim, K. Kim, S. Lee, G. Song, M.S. Jung, K.T. Lee, Thermal runaway behavior of $\text{Li}_6\text{PS}_5\text{Cl}$ solid electrolytes for $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ and LiFePO_4 in all-solid-state batteries, *Chem. Mater.* 34 (2022) 9159–9171, <https://doi.org/10.1021/acs.chemmater.2c02106>.
- [42] F. Walther, S. Randau, Y. Schneider, J. Sann, M. Rohnke, F.H. Richter, W.G. Zeier, J. Janek, Influence of carbon additives on the decomposition pathways in cathodes of lithium thiophosphate-based all-solid-state batteries, *Chem. Mater.* 32 (2020) 6123–6136, <https://doi.org/10.1021/acs.chemmater.0c01825>.
- [43] F. Li, X.B. Cheng, L.L. Lu, Y.C. Yin, J.D. Luo, G.X. Lu, Y.F. Meng, H.S. Mo, T. Tian, J. T. Yang, W. Wen, Z.P. Liu, G.Z. Zhang, C. Shang, H.B. Yao, Stable all-solid-state lithium metal batteries enabled by machine learning simulation designed halide electrolytes, *Nano Lett.* 22 (2022) 2461–2469, <https://doi.org/10.1021/acs.nanolett.2c00187>.